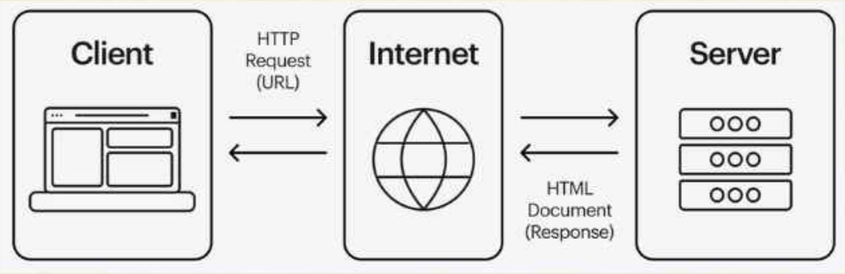
Web Technology

WEB ESSENTIALS AND MARK-UP LANGUAGE HTML

unit 1

WEB TECHNOLOGY & INTERNET

- Web technology acts as an interface between a web client (browser) and a web server.
- It includes:
- Markup languages (HTML, XML)
- Client-side scripting languages (JavaScript, CSS)
- Server-side technologies (ASP, JSP, PHP, Java Servlets)
- Online presentation tools, Java applets, video tools, etc.
- It helps in creating, designing, and managing websites and web applications.



Concept of Internet

- The Internet is a global network connecting millions of computers worldwide.
- It is called a network of networks.
- It allows users to:
- Share resources
- Communicate globally
- Access information
- To use the Internet, a computer must connect through an Internet Service Provider (ISP) using modem, DSL, or other connections.

CONCEPT OF WORLD WIDE WEB (WWW)

Definition of WWW

- The World Wide Web (WWW) is a collection of software and protocols used to access resources over the Internet.
- It is an information system where documents are interlinked using hyperlinks.

Features of WWW

- Information is stored in the form of web pages.
- Web pages are accessed using web browsers.
- Web pages may contain:
  - Text
  - Images
  - Audio
  - Video
  - Graphics
- Users can navigate between pages using hyperlinks.

History of WWW

- The concept of WWW was introduced by Tim Berners-Lee.
- He developed it in 1980 at CERN, Switzerland.
- He used hypertext to link documents together.

BASIC INTERNET PROTOCOLS

File Transfer Protocol (FTP)

- Used to transfer files between client and server.
- Helps in downloading and uploading files.
- Uses two connections:
  - One for data transfer.
  - One for control/commands.
- Separation of data and control makes FTP more efficient.

Hyper Text Transfer Protocol (HTTP)

- A request/response communication protocol.
- Used to transfer web pages and resources on the WWW.
- Works through TCP/IP.
- Called a stateless protocol because it does not remember previous requests.

Simple Network Management Protocol (SNMP)

- Used to manage and monitor network devices.
- Helps network administrators diagnose problems.
- Has two main components:
  - Supervisor (network management system)
  - Agents (installed on network devices like routers, switches)

Simple Mail Transfer Protocol (SMTP)

- Used to send emails to remote servers.
- Asynchronous protocol (allows delayed message delivery).
- Works with Mail Transfer Agent (MTA) and User Agent.

BASIC INTERNET PROTOCOLS

Post Office Protocol Version 3 (POP3)

- Used by email clients to receive emails.
- Works at the receiver's end.
- Requires authentication (username & password).
- Establishes TCP connection with mail server.

User Datagram Protocol (UDP)

- Connectionless protocol.
- No error detection.
- Faster but unreliable compared to TCP.
- Used for quick data transmission.

Transmission Control Protocol (TCP)

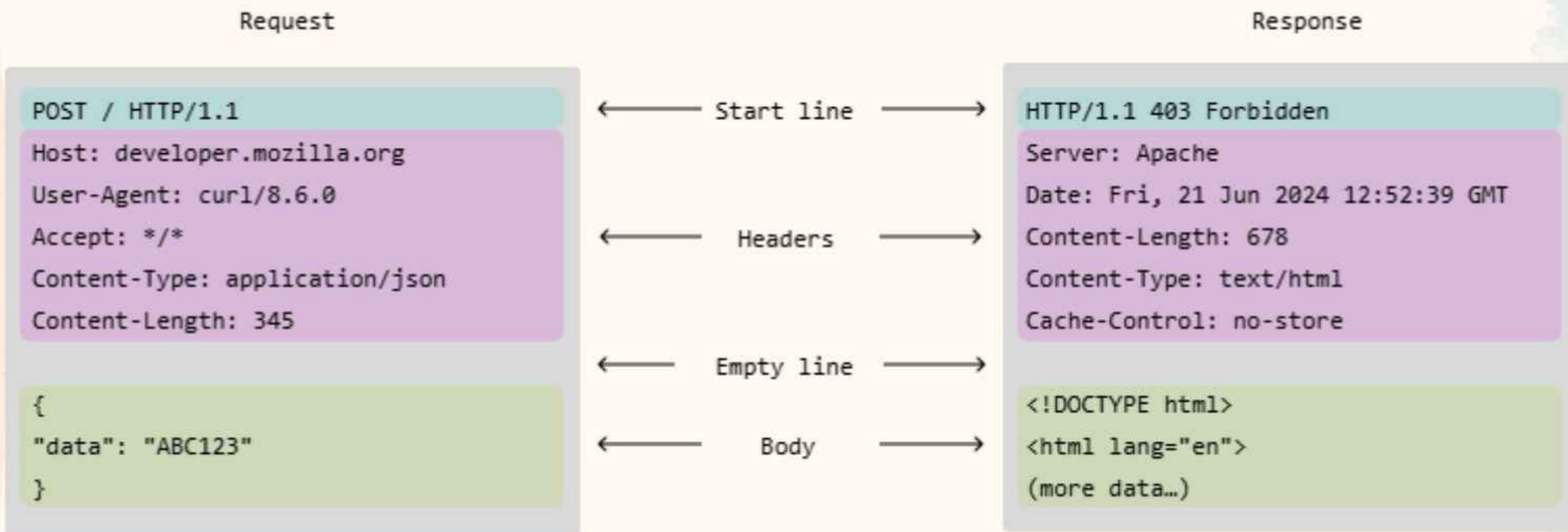
- Ensures safe and reliable data delivery.
- Performs:
  - Error detection
  - Correct sequencing
  - Acknowledgement of data
- Connection-oriented protocol.
- Breaks data into packets and reassembles them properly.

Internet Protocol (IP)

- Network layer protocol.
- Provides unique addressing to computers.
- Enables communication between devices on the Internet.

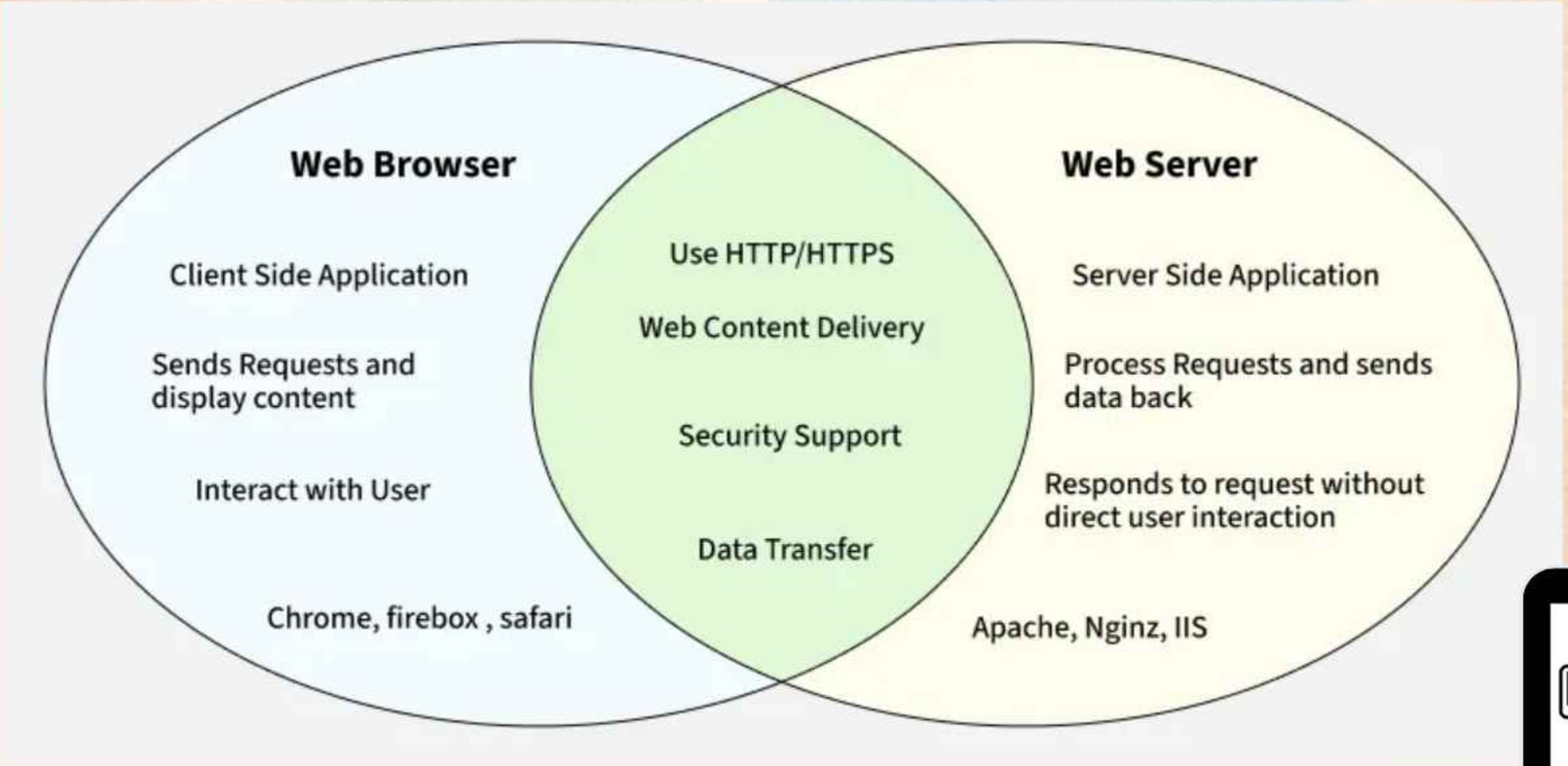
HTTP REQUEST MESSAGE & HTTP RESPONSE MESSAGE

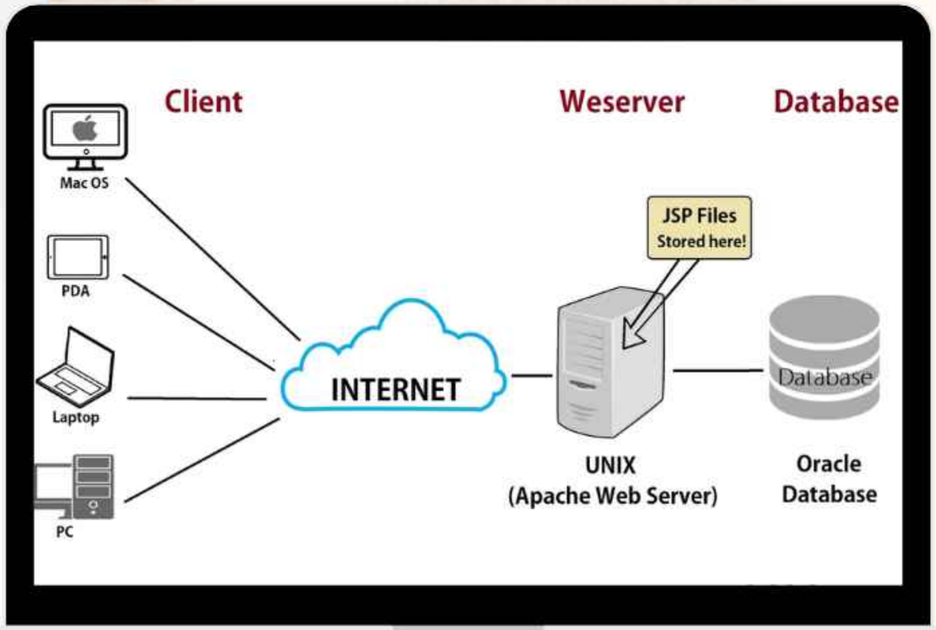
An HTTP request message is sent by the client (browser) to the server to request a resource (web page, image, file, etc.).



An HTTP response message is sent by the server to the client after processing the request.

Web clients AND Web servers

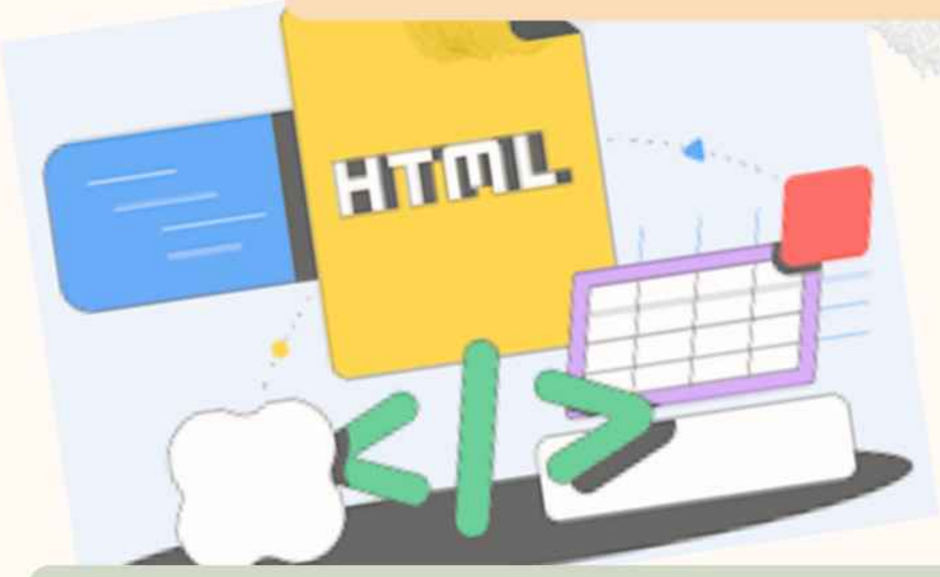




What are desirable features for good website?

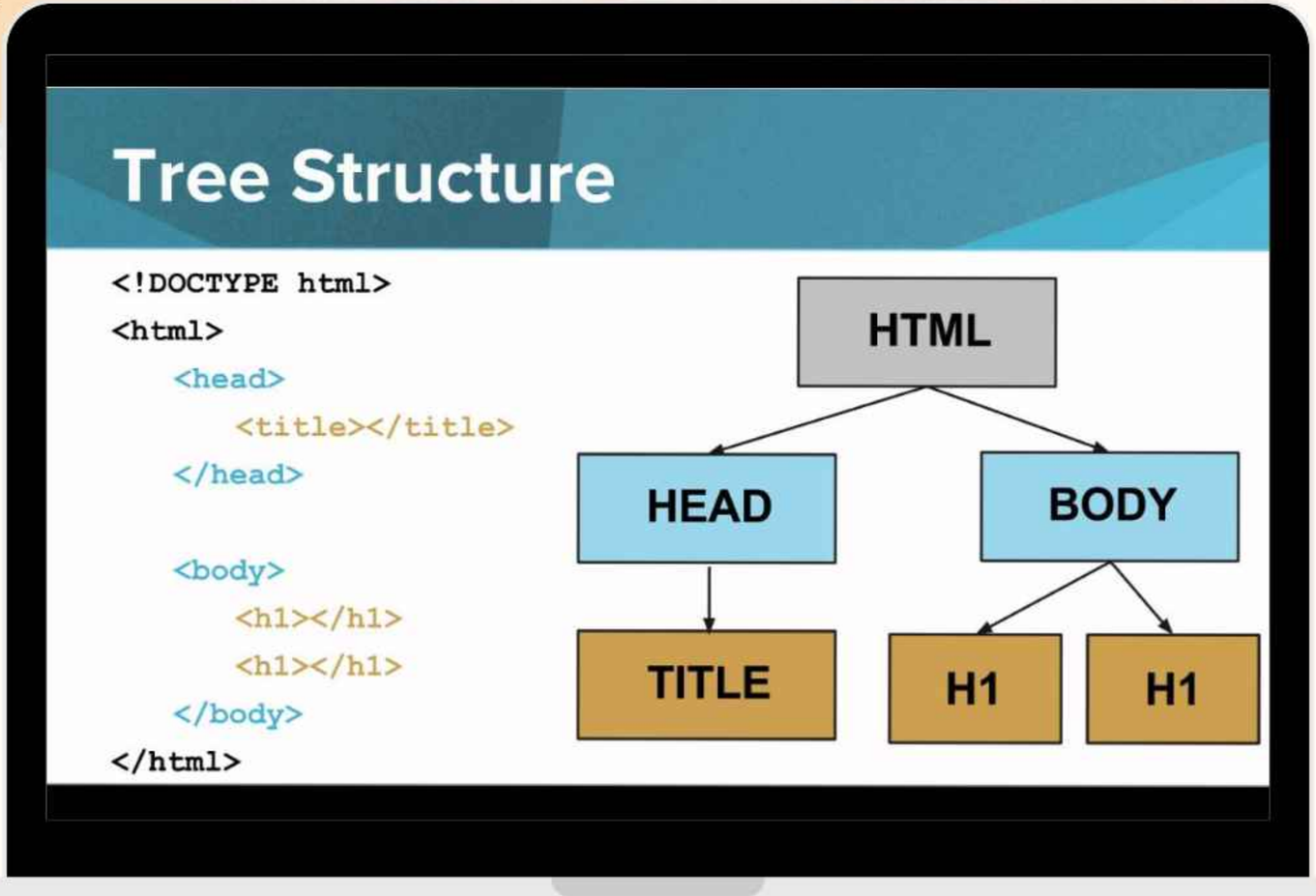
HTML – Introduction

- HTML stands for HyperText Markup Language.
- It is the standard markup language used to create web pages.
- It defines the structure of a web page.
- HTML tells the browser how to display content.



Uses of HTML

- Creating web pages.
- Structuring content (headings, paragraphs, lists).
- Adding links, images, tables, and forms.



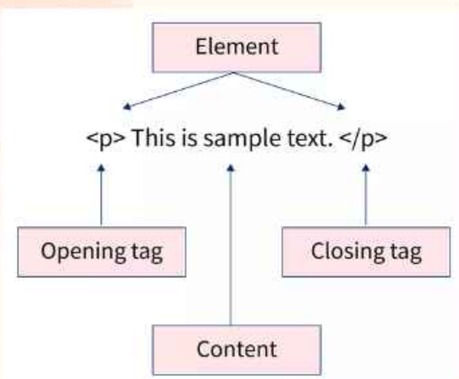
HTML ELEMENTS

An HTML element is the basic building block of an HTML document.
It defines the structure and content of a web page.

Structure of an HTML Element:
An HTML element usually consists of:

1. Opening Tag
2. Content
3. Closing Tag

Example:



HTML ELEMENTS

CONTAINER ELEMENTS

- HAVE OPENING AND CLOSING TAGS.
- EXAMPLE: <H1>, <P>, <DIV>

EMPTY ELEMENTS

- DO NOT HAVE CLOSING TAGS.
- EXAMPLE: <BR>, <HR>, <IMG>

HTML ELEMENTS

Headings (<h1> to <h6>)

- Used to define headings on a web page.
- There are 6 levels of headings:
  - <h1> – Largest heading
  - <h6> – Smallest heading
- Helps in organizing content properly.
- Important for SEO and readability.

Example:

```
<h1>Main Heading</h1>
<h2>Sub Heading</h2>
```

Paragraph (<p>)

- Used to define a paragraph of text.
- Automatically adds space before and after the paragraph.

Example:

```
<p>This is a paragraph in HTML.</p>
```

Line Break (<br>)

- Used to break a line without starting a new paragraph.
- It is an empty tag (no closing tag).

Example:

```
This is first line.<br>
This is second line.
```

Colors in HTML

- Used to make web pages attractive.
- Can be applied to:
  - Text
  - Background
  - Borders
- Defined using:
  - Color names (red, blue)
  - HEX values (#FF0000)
  - RGB values (rgb(255,0,0))

Example:

```
<p style="color:blue;">Blue Text</p>
```

HTML ELEMENTS

Fonts in HTML

- Used to change text style and appearance.
- Controlled using CSS:
  - Font-family
  - Font-size
  - Font-style
  - Font-weight

Example:

```
<p style="font-family:Arial; font-size:20px;">
Hello World
</p>
```

Links in HTML

- Created using <a> (anchor) tag.
- Used to navigate between web pages.
- href attribute specifies the URL.

Example:

```
<a href="https://www.google.com">Visit Google</a>
```

Frames in HTML

- Used to divide a browser window into multiple sections.
- Each section can load a different HTML document.
- <frame> and <frameset> were used earlier.
- Now replaced by <iframe> (Inline Frame).

Example:

```
<iframe src="page.html"></iframe>
```

Lists in HTML

Used to display items in an organized way.

- Ordered List (<ol>) – Numbered list
- Unordered List (<ul>) – Bulleted list
- List Item (<li>) – Defines each item

Example:

```
<ul>
  <li>Apple</li>
  <li>Banana</li>
</ul>
```

HTML ELEMENTS

Images in HTML

Used to display images on a web page.

- <img> tag is used.
- src attribute defines image path.
- alt attribute provides alternative text.

Example:

```
<img src="image.jpg" alt="Sample Image">
```

Tables in HTML

Used to display data in rows and columns.

- <table> – Defines table
- <tr> – Table row
- <th> – Table heading
- <td> – Table data

Example:

```
<table border="1">
  <tr>
    <th>Name</th>
    <th>Age</th>
  </tr>
  <tr>
    <td>John</td>
    <td>20</td>
  </tr>
</table>
```

Forms in HTML

Used to collect user input.

- <form> – Defines form
- <input> – Input fields
- <label> – Field label
- <button> – Submit button

Example:

```
<form>
  Name: <input type="text">
  <button type="submit">Submit</button>
</form>
```

Welcome to HTML Demo

This is a Sub Heading

This is a paragraph example.
This line is after line break.

This text is red and larger in size.

Visit Website:

Click here to open Google

Fruits List:

- Apple
- Banana
- Mango

Subjects (Ordered List):

1. HTML
2. CSS
3. JavaScript

Student Table:

Name	Age
Vrunda	20
Rahul	21

Sample Image:

Sample Image

Contact Form:

Name:

Email:

Password:

Submit

```html
<!DOCTYPE html>
<html>
<head>
    <title>All HTML Elements Example</title>
</head>

<body style="background-color: #f2f2f2; font-family: Arial;">

    <!-- Heading -->
    <h1 style="color: blue;">Welcome to HTML Demo</h1>
    <h2>This is a Sub Heading</h2>

    <!-- Paragraph and Line Break -->
    <p>
        This is a paragraph example.<br>
        This line is after line break.
    </p>

    <!-- Colored Text -->
    <p style="color: red; font-size: 18px;">
        This text is red and larger in size.
    </p>

    <!-- Link -->
    <h3>Visit Website:</h3>
    <a href="https://www.google.com" target="_blank">
        Click here to open Google
    </a>

    <hr>

    <!-- List -->
    <h3>Fruits List:</h3>
    <ul>
        <li>Apple</li>
        <li>Banana</li>
        <li>Mango</li>
```

Iframe Example:

```html
<table border="1">
    <tr>
        <th>Name</th>
        <th>Age</th>
    </tr>
    <tr>
        <td>Vrunda</td>
        <td>20</td>
    </tr>
    <tr>
        <td>Rahul</td>
        <td>21</td>
    </tr>
</table>

<hr>

<!-- Image -->
<h3>Sample Image:</h3>
<img src="https://via.placeholder.com/150" alt="Sample Image">

<hr>

<!-- Form -->
<h3>Contact Form:</h3>
<form>
    Name: <input type="text">><br><br>
    Email: <input type="email"><br><br>
    Password: <input type="password"><br><br>
    <button type="submit">Submit</button>
</form>

<hr>
```

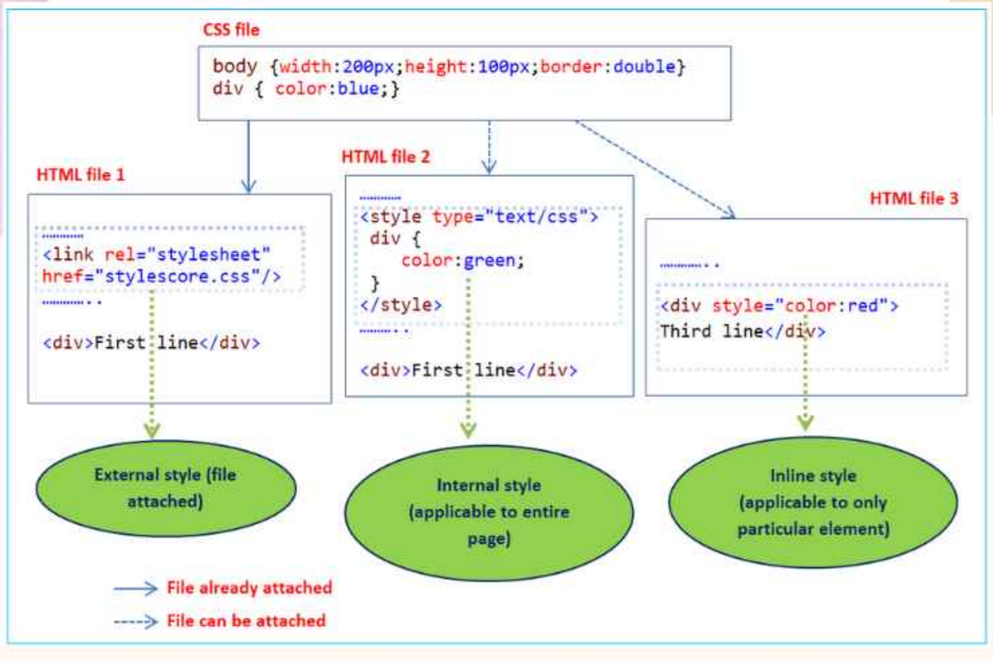
HTML	HTML5
Older version of HyperText Markup Language.	Latest version of HTML.
Limited multimedia support (needs plugins like Flash).	Built-in support for audio and video (<audio>, <video>).
No semantic elements like <article>, <section>.	Introduced semantic elements for better structure.
Does not support canvas for graphics.	Supports <canvas> for drawing graphics.
No built-in local storage.	Supports local storage (localStorage, sessionStorage).
Less mobile-friendly.	Designed to be mobile-friendly.
Simpler form controls.	New input types like email, date, number, range.
Uses long DOCTYPE declaration.	Simple DOCTYPE: <!DOCTYPE html>

CSS: Introduction to Style Sheet & Features

- CSS stands for Cascading Style Sheets.
- It is used to style and design HTML web pages.
- CSS controls the appearance of elements like:
  - Colors
  - Fonts
  - Layout
  - Spacing
- It separates content (HTML) from design (CSS).

Features of CSS

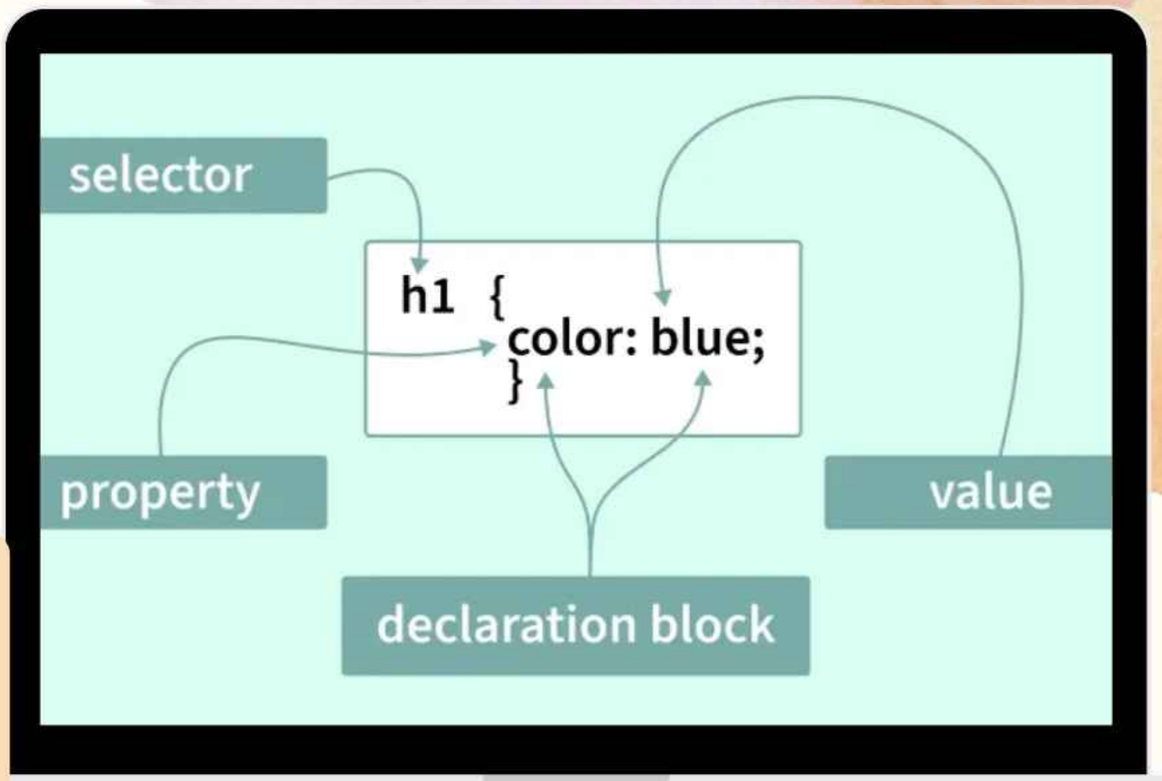
- Improves website design and presentation.
- Reduces code repetition.
- Makes website maintenance easy.
- Provides responsive design.
- Supports animations and transitions.
- Faster page loading (with external CSS).
- Better control over layout and positioning.



CSS: What is CSS Syntax?

CSS syntax defines how style rules are written to apply styles to HTML elements.

```
selector {
    property: value;
}
```



Important Points:

- Each declaration ends with a semicolon ;
- Multiple properties are written inside { }
- CSS is case-insensitive.
- Comments in CSS are written as:

```
/* This is a comment */
```

Basic Types of CSS Selectors

1) Element Selector

```
p { color: blue; }
```

2) ID Selector

```
#header { background-color: yellow; }
```

3) Class Selector

```
.box { color: red; }
```

4) Universal Selector

```
* { margin: 0; }
```

5) Grouping Selector

```
h1, h2, p { color: green; }
```

6) Attribute Selector

```
input[type="text"] {
  border: 2px solid green; }
}
```

CSS Cascading

When multiple CSS rules apply to the same element, CSS decides the final style using the following priority order.

priority order:

- **Origin and Importance**

  !important has highest priority.

  Inline styles override internal and external styles.

- **Cascade Layers**

  Styles are applied based on defined layers.

  Higher layers override lower layers.

- **Selector Specificity**

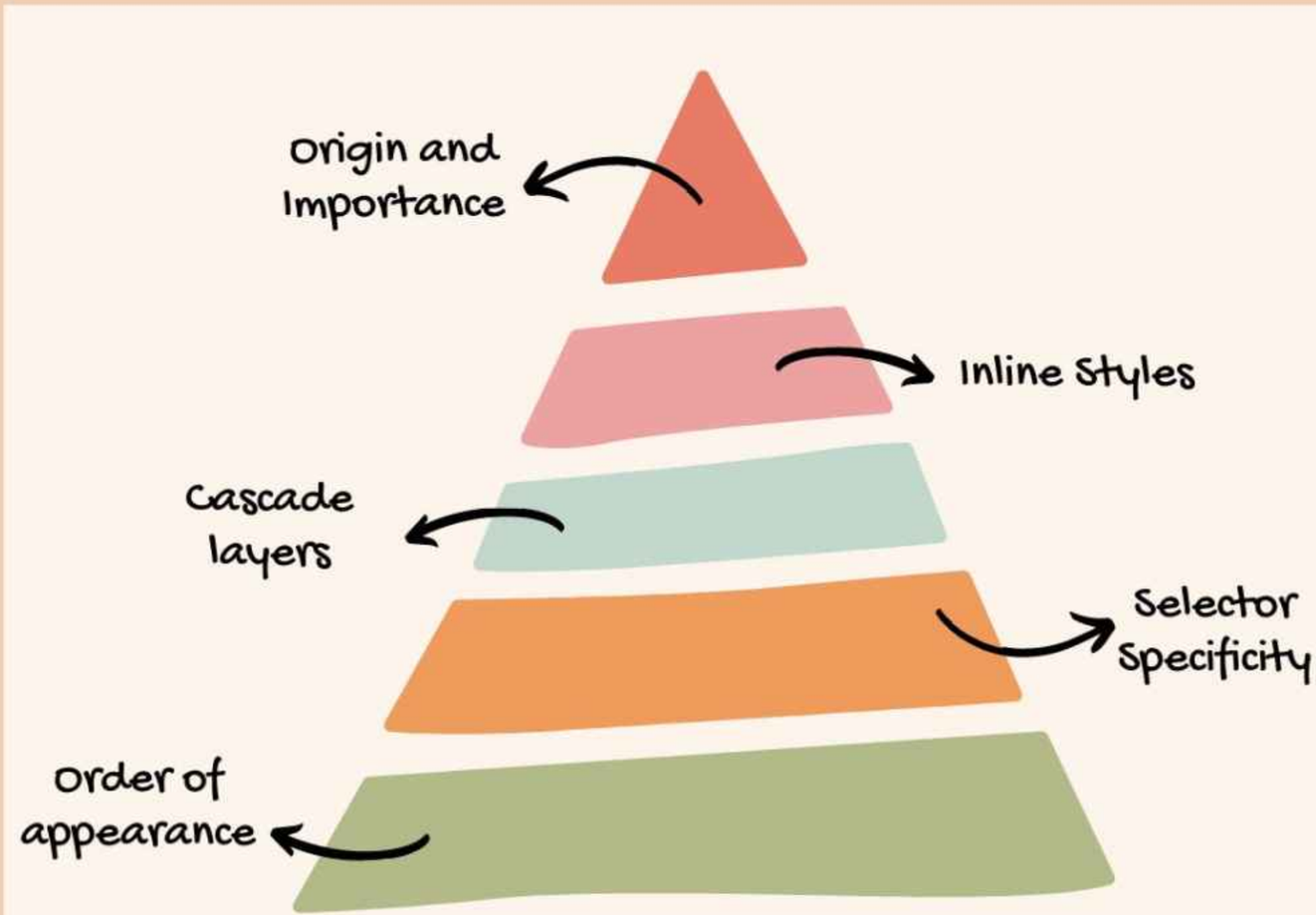
  ID selector > Class selector > Element selector.

  More specific selector gets higher priority.

- **Order of Appearance**

  If specificity is same, the rule written later is applied.

CSS follows top-to-bottom and left-to-right order.



CSS Cascading

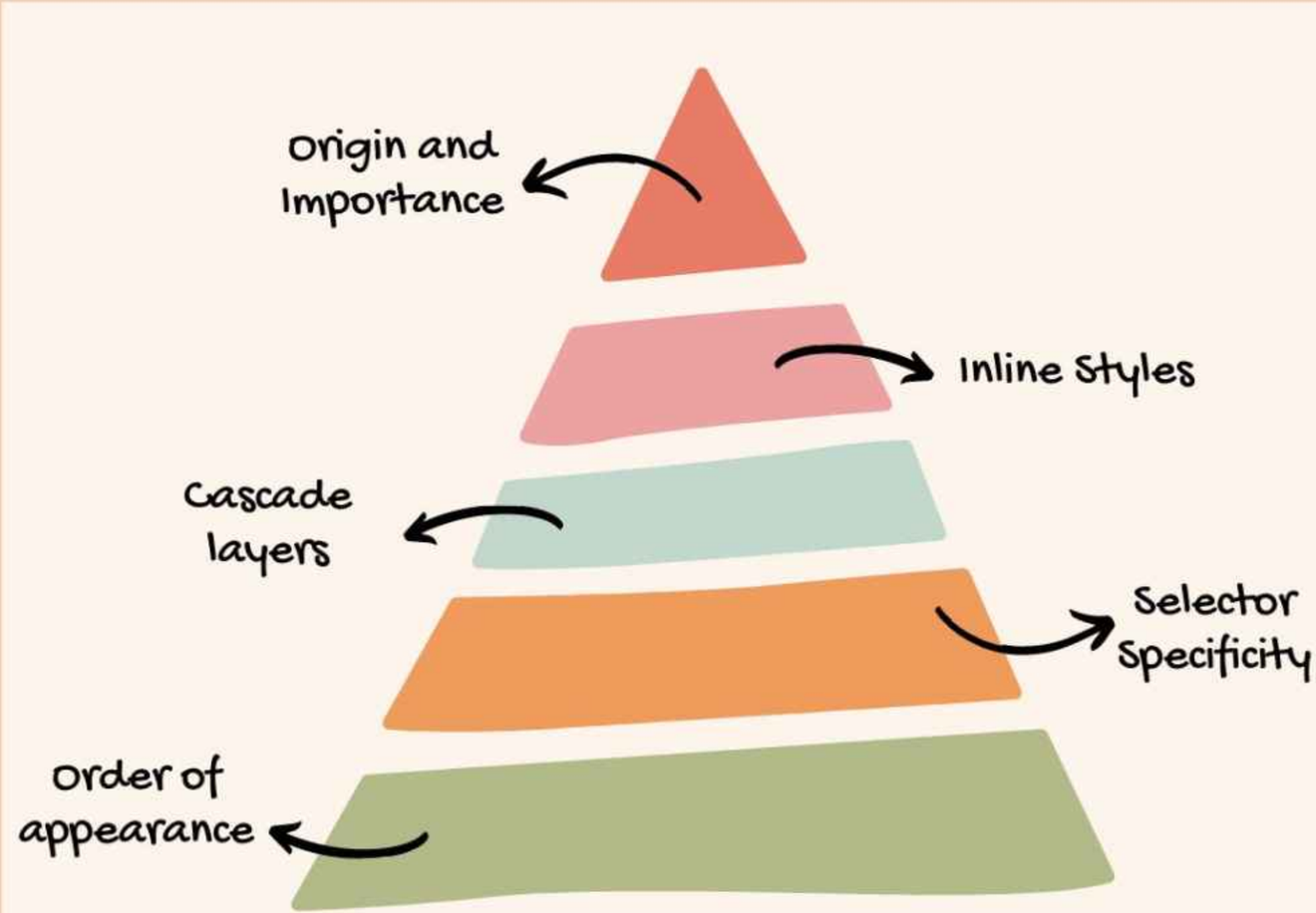
When multiple CSS rules apply to the same element, CSS decides the final style using the following priority order.

```
<!DOCTYPE html>
<html>
<head>
<style>
p { color: blue; }          /* Element selector */
.text { color: green; }     /* Class selector */
#para { color: orange; }    /* ID selector */
</style>
</head>

<body>

<p id="para" class="text" style="color: red;">
 Hello CSS
</p>

</body>
</html>
```



TEXT PROPERTIES

CSS Font Properties

Font Family

- Used to specify the type of font.
- Example: Arial, Times New Roman, Garamond
- Generic families:
  - Serif – Times New Roman
  - Sans-serif – Arial
  - Monospace – Courier
  - Cursive – Script
  - Fantasy – Decorative fonts

```
h1 { font-family: Arial; }
```

Example: 'Times New Roman'

Font Size

- Specifies size of text
- Units:
  - pt (points)
  - px (pixels)
  - % (percentage)
  - relative values: small, medium, large

```
h1 { font-size: 20px; }
```

Font Variant

- Used to display text in small capital letters

Values:

- normal
- small-caps

```
h1 { font-variant: small-caps; }
```

Font Style

Values:

- normal
- italic
- oblique

```
h1 { font-style: italic; }
```

Font Weight

Values:

- normal
- bold
- bolder
- lighter

```
p { font-weight: bold; }
```

TEXT PROPERTIES

CSS Text Properties

Font Shorthand

- Multiple font properties can be written in one line.

```
p {
    font: italic bold 20px Arial;
}
```

Text Decoration

Adds special effects to text.

Values:

- underline
- overline
- line-through
- none

```
p { text-decoration: underline; }
```

Text Alignment

Aligns text on webpage.

Values:

- left
- right
- center

```
h1 { text-align: center; }
```

Text Indent

Adds space before first line of paragraph.

```
p { text-indent: 50px; }
```

Multi-Color Text Example

```
<p>
  <span style="color:red;">Hello</span>
  <span style="color:blue;">World</span>
</p>
```

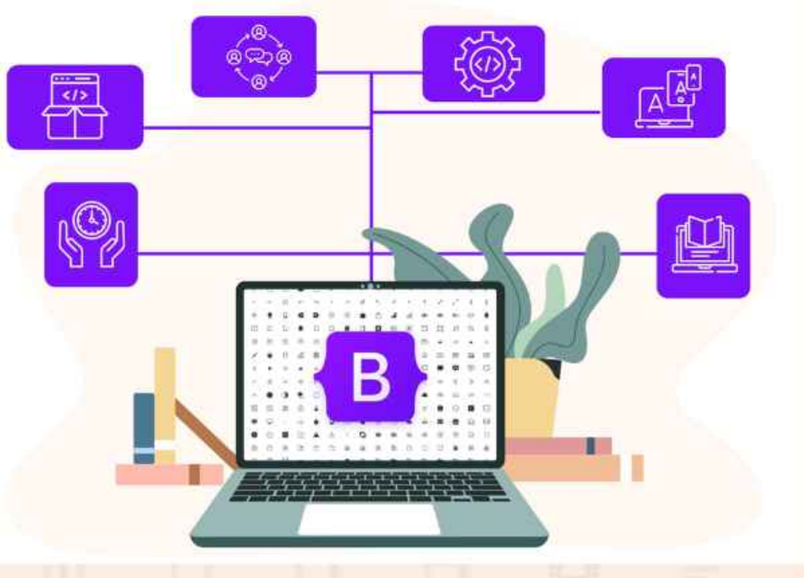
INTRODUCTION TO BOOTSTRAP

What is Bootstrap?

- Bootstrap is a front-end framework.
- It is used to design responsive and mobile-first websites.
- Developed by Twitter.

It includes:

- CSS
- JavaScript
- Pre-designed components



Why Use Bootstrap?

- Saves development time
- Responsive design support
- Easy to use
- Cross-browser compatibility
- Large community support

How to Include Bootstrap?
Using CDN:

```
<link href="https://cdn.jsdelivr.net/npm/bootstrap@5.3.0/dist/css/bootstrap.min.css" rel="stylesheet"
```

BOOTSTRAP GRID SYSTEM

Bootstrap Grid System is a powerful layout system used to create responsive web designs.

- Based on 12 column structure
- Helps in arranging content easily
- Automatically adjusts according to screen size
- Mobile-first approach

Bootstrap provides different classes for different screen sizes:

Class	Screen Size
xs (<576px)	Portrait Mobile
sm (≥576px)	Landscape Mobile
md (≥768px)	Tablets
lg (≥992px)	Small Laptops
xl (≥1200px)	Large Desktops

```
<div class"col-xs-2 col-sm-6 col-md-4 col-lg-5 col-xl-6'">
```

COMPONENTS OF BOOTSTRAP GRID SYSTEM

Containers

Bootstrap requires a container to wrap the content.

Types:

- container → fixed width
- container-fluid → full width

Example:

```
<div class="container">
  Content Here
</div>
```

Rows

- Rows must be inside containers.
- Used to create horizontal groups.

Example:

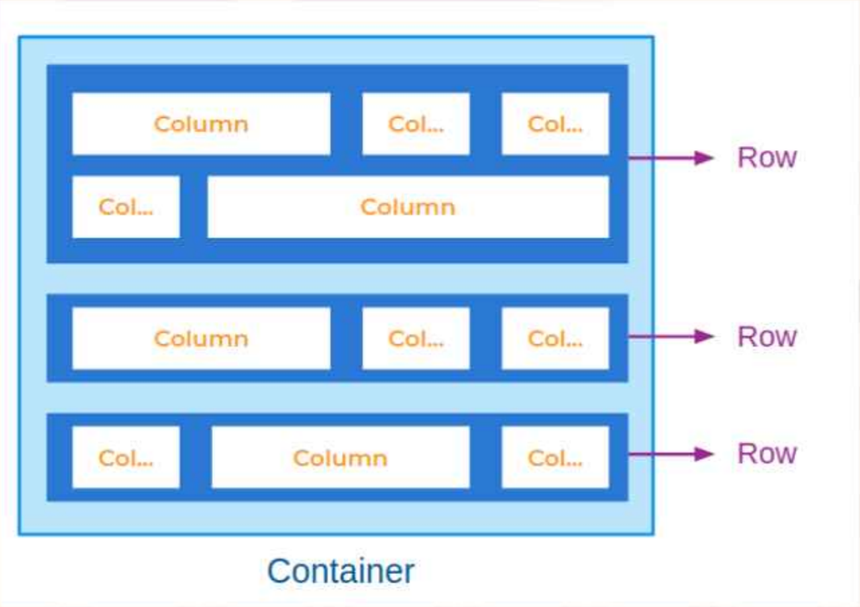
```
<div class="row">
  <div class="col-md-6">Row 1</div>
  <div class="col-md-6">Row 2</div>
</div>
```

Columns

- Columns are created inside rows.
- Total columns in one row must be 12.

Example:

```
<div class="col-sm-4">Column 1</div>
<div class="col-sm-4">Column 2</div>
<div class="col-sm-4">Column 3</div>
```



Thanks for watching

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